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Campus Network Design and Packet Trace Simulator

Campus Packet Lab

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GitHub: https://github.com/AsHx-BiT/Campus-Network-

Problem and Objectives

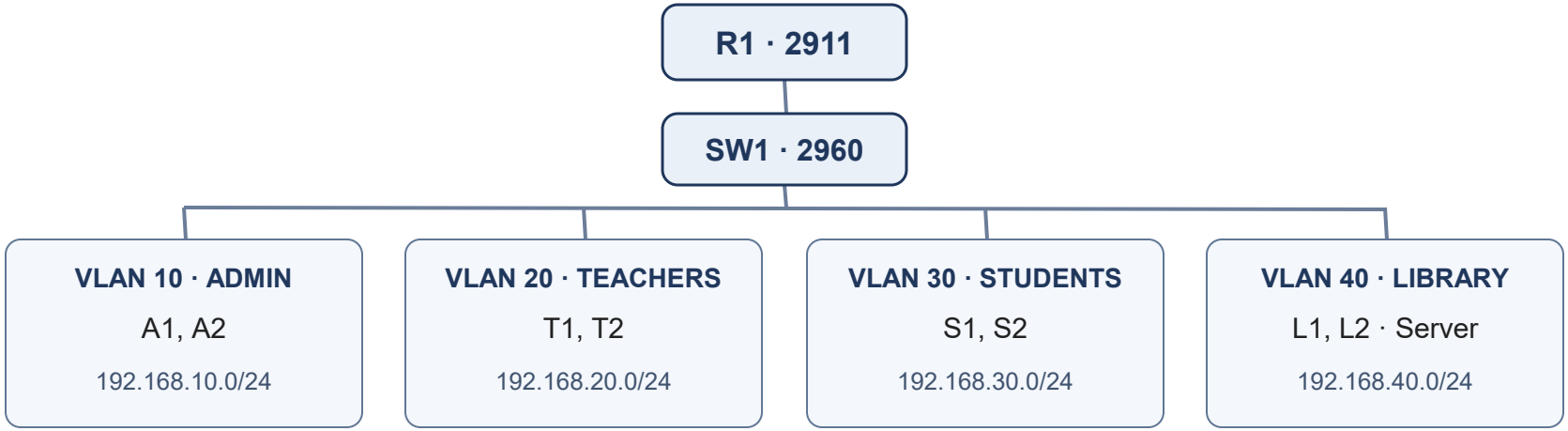
Problem

A campus needs separate networks for administration, teachers, students and library users.
The design should let each group communicate while keeping traffic organized.

Objectives

1. Divide the campus LAN into four VLANs.
2. Route between VLANs with one router trunk.
3. Provide DHCP, DNS and a school web page, then trace sample packets.

Proposed Design and Tools



Tools: Cisco Packet Tracer; companion dark-theme web simulator

Protocols: 802.1Q trunk, IPv4, DHCP, DNS and HTTP

Server: 192.168.40.10 | Gateway: .1 in each subnet

Implementation and Working

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!  Repeat for VLANs 20, 30 and 40
```

Implementation: VLAN access ports, router subinterfaces and DHCP pools. Server .10 runs DNS and HTTP.
Working: A VLAN tag reaches R1; R1 routes to the destination VLAN and SW1 forwards the frame.
Fix applied: enable R1 Gi0/0; reserve the server IP from DHCP.

Testing and Results

A1 (VLAN 10) → SW1 → R1 → SW1 → Server (VLAN 40)
TCP destination port 80 | 4 network hops | TTL 64 → 63

Test: simulator model, A1 → Server using TCP port 80.
Expected: cross-VLAN traffic is routed by R1 to the server subnet.
Observed: path includes R1; modeled TTL decreases by one.
Also checked: same-VLAN path bypasses R1; DNS uses server 192.168.40.10.

Working Demonstration

Function to demonstrate

Show how a packet moves from a client in one VLAN to the campus server in another VLAN.

1. Open the completed .pkt in Cisco Packet Tracer and show the topology.
2. In the local website, choose A1 → Server, TCP port 80, then run Trace.
3. Compare the route and hop details with the VLAN layout.

Success: the trace crosses R1 between VLANs and reaches the server.

Conclusion and Contributions

Outcome: four VLANs, DHCP, DNS and a campus web server are configured in the project.
Use: demonstrate department separation and inter-VLAN routing.
Limitation: website traces are modeled; live Packet Tracer traffic was not tested here.
Future: add more campus services and security rules.

Team contribution: Ajinkya Katke, Tattwam Samui, Akshat Pandey, Smith Hajgude and Shlok Pandey — project team.

AI use: helped build the companion simulator and configuration guidance; details checked against the project files.

References

[1] Cisco Networking Academy, “Cisco Packet Tracer,” https://www.netacad.com/catalogs/learn/cisco-packet-tracer.

[2] Cisco, “Configure Inter VLAN Routing with the Use of an External Router,” https://www.cisco.com/c/en/us/support/docs/lan-switching/inter-vlan-routing/14976-50.html.

[3] R. Droms, “Dynamic Host Configuration Protocol,” RFC 2131, IETF, Mar. 1997.

Project evidence: Group_project_completed.pkt and the Campus Packet Lab local website.